Week 02: two kinds of work

Programming task

Session 2: three short Python tasks.
Predict, change and repair.
Session 3: separate application.

Research task

Share Research 1 in Session 1.
After class: Research 2.
Discuss in Week 3 · proposed 5%.

Use this guide with the same Week notebook; do not create a second set of weekly tasks.

Programming practice

Comparisons, Boolean values and branches
Predict → run → change → repair → explain

Three core tasks: understand the Python

Predict a branch

Trace enabled and score >= 20. Predict the selected branch.

Change and observe

Try 19 / 20 with True, then 21 with False. Explain changes.

Repair the condition

Replace the wrong or. False / 21 must stay waiting.

Open Week_02.ipynb → Session 2. Keep a prediction, result and explanation for each task.

Core 1–2: predict, then change one thing

```
01 score = 20
02 enabled = True
03 print(score >= 20)
04 if enabled and score >= 20:
05     print("ready")
06 else:
07     print("waiting")
08 print("finished")
```

1. Trace the Boolean condition and selected block.
2. Change inputs: (19, True), (20, True), (21, False).
3. Expected branches: waiting, ready, waiting.

Use the complete Core 1 and Core 2 starters in the notebook; explain each changed result.

Core 3: find and repair one mistake

```
score = 21
enabled = False
if enabled or score >= 20:  # Deliberate Boolean bug.
    print("ready")
else:
    print("waiting")
# Repair and retest (19, True), (20, True), (21, False).
```

1. Rule: ready requires BOTH conditions.
2. False / 21 must print waiting.
3. Repair the operator; retest 19, 20 and the disabled case.

Edit the matching Core 3 cell. Keep the wrong result, your repair and a retest.

Done: explain the concept, then save

Step	What to do / keep
Predict	Explain comparison → Boolean → selected branch.
Change	Keep results for (19, True), (20, True), (21, False).
Repair	Explain why or was wrong; preserve all three results.
Optional	Try 0 and 21 with both Boolean values.
Save	Keep code and three records in Week_02.ipynb.

After practice, explain how comparisons and Boolean operators choose a branch.

Project application

Continue in the same notebook after the Python practice.

Apply the idea: review before workload

Step	What to do / keep
Open	Project studio → Independent lab: combine review and workload.
Edit	Use total_portions and reviewed in the lab starter.
Run / check	False: hold for 19/20/21. True: standard/team/team.
Write / save	Six predicted/actual labels, one bug case and a revision.

Use the project rule at 20; ready/waiting belonged only to the Python exercise.

Research functions and testing

What to research

Find out what a function does and why we test it.

Possible directions

Look at one small function: its inputs, returned value and a test that could catch a believable mistake.

A short summary example

A minutes-to-seconds function should return 120 for 2. Adding 60 would give 62, so this test catches that mistake.

Discuss in Week 3 · about 15–20 minutes · summarize in your own words and cite your sources.